Supplement to "Who Is Really on the Frontier? Confidence Sets for Cost-Accuracy Pareto Leaderboards"

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Every table here is generated from the result files by `src/make_supplement.py`; the numbers are the ones the main text uses. System names are the agent-and-model part of HAL's run names; a dagger marks a board whose costs are run totals spread evenly over the tasks and treated as fixed.

S1. Every system on every board

Cost per task is in dollars under one price table with each model call counted once. "Frontier" marks the observed frontier, "In set" the frontier confidence set at the 5% level, "Best set" the confidence set of the most accurate system. The rival lead is the strongest cheaper-or-equal system's accuracy minus this system's, in paired standard errors; a system leaves the frontier set when its rival lead exceeds the board's simultaneous margin.

Table S1: AssistantBench: 32 systems on 33 tasks, cheapest first. Uncosted: tasks with an outcome but no logged model call, priced at the run's mean costed task. In set (fixed): the frontier set with every cost fixed at its mean. Rival lead: how far the strongest cheaper-or-equal system sits above this one, in paired standard errors (negative when this system is ahead); the simultaneous margin for this board is 3.00.

System	$/task	Uncosted	Acc. %	Frontier	In set	In set (fixed)	Undom. fre
HAL Generalist Agent (gemini-2.0-flash)	0.021	0	3.0	yes	yes	yes	0.84
Assistantbench Browser Agent (gemini-2.0-flas	0.028	0	0.0		yes	yes	0.08
Hal Generalist Agent (o4-mini-2025-04-16 low)	0.029	0	3.0		yes	yes	0.15
Assistantbench Browser Agent (deepseek-ai/Dee (1)	0.041	0	0.0		yes	yes	0.00
HAL Generalist Agent (deepseek-ai/DeepSeek-V3)	0.060	0	12.1	yes	yes	yes	0.84
Assistantbench Browser Agent (o4-mini-2025-04 (1)	0.062	0	6.1		yes	yes	0.33
Hal Generalist Agent (o3-2025-04-16)	0.073	0	0.0		yes	yes	0.00
Hal Generalist Agent (o4-mini-2025-04-16)	0.080	0	3.0		yes	yes	0.00
Assistantbench Browser Agent (gpt-4.1-2025-04	0.084	0	6.1		yes	yes	0.10
Assistantbench Browser Agent (o3-2025-04-16)	0.089	0	6.1		yes	yes	0.10
Assistantbench Browser Agent (o4-mini-2025-04 (2)	0.094	0	6.1		yes	yes	0.03
Assistantbench Browser Agent (deepseek-ai/Dee (2)	0.101	0	0.0		yes	yes	0.00
Hal Generalist Agent (gpt-4.1-2025-04-14)	0.104	0	9.1		yes	yes	0.09
Assistantbench Browser Agent (o4-mini-2025-04 (3)	0.120	0	6.1		yes	yes	0.00
Hal Generalist Agent (o4-mini-2025-04-16 high)	0.127	0	6.1		yes	yes	0.01
Assistantbench Browser Agent (gpt-5) (1)	0.293	0	0.0		yes	yes	0.00
HAL Generalist Agent (claude-3-7-sonnet-20250 (1)	0.294	0	0.0		yes	yes	0.00
Assistantbench Browser Agent (gpt-5) (2)	0.295	0	0.0		yes	yes	0.00
HAL Generalist Agent (claude-3-7-sonnet-20250 (2)	0.297	0	3.0		yes	yes	0.00
HAL Generalist Agent (claude-3-7-sonnet-20250 (3)	0.320	0	3.0		yes	yes	0.00
HAL Generalist Agent (deepseek-ai/DeepSeek-R1)	0.370	0	0.0		yes	yes	0.00
Assistantbench Browser Agent (claude-3-7-sonn (1)	0.434	0	3.0		yes	yes	0.02
Assistantbench Browser Agent (claude-3-7-sonn (2)	0.451	0	9.1		yes	yes	0.14
Assistantbench Browser Agent (claude-3-7-sonn (3)	0.489	0	0.0		yes	yes	0.00
Assistantbench Browser Agent (claude-sonnet-4 (1)	0.574	0	3.0		yes	yes	0.00
Assistantbench Browser Agent (claude-sonnet-4 (2)	0.756	0	6.1		yes	yes	0.03
Assistantbench Browser Agent (deepseek-R1)	1.107	0	0.0		yes	yes	0.00
Assistantbench Browser Agent (claude-opus-4.1 (1)	2.983	0	3.0		yes	yes	0.00

System	$/task	Uncosted	Acc. %	Frontier	In set	In set (fixed)	Undom. fre
HAL Generalist Agent (claude-opus-4.1-2025051 (1)	3.287	0	6.1		yes	yes	0.03
HAL Generalist Agent (claude-opus-4.1-2025051 (2)	3.412	0	6.1		yes	yes	0.03
HAL Generalist Agent (claude-opus-4.1 high)	4.394	0	15.2	yes	yes	yes	0.42
Assistantbench Browser Agent (claude-opus-4.1 (2)	10.094	0	0.0		yes	yes	0.00

Table S2: ColBench backend programming: 22 systems on 1000 tasks, cheapest first. Uncosted: tasks with an outcome but no logged model call, priced at the run's mean costed task. In set (fixed): the frontier set with every cost fixed at its mean. Rival lead: how far the strongest cheaper-or-equal system sits above this one, in paired standard errors (negative when this system is ahead); the simultaneous margin for this board is 2.59.

System	$/task	Uncosted	Acc. %	Frontier	In set	In set (fixed)	Undom
colbench_text_deepseekv3	0.002	202	21.0	yes	yes	yes	1.0
colbench_backend_programming colbench_example (1)	0.002	0	24.5	yes	yes	yes	1.0
colbench_text_gemmini2flash	0.006	0	32.9	yes	yes	yes	0.9
colbench_backend_programming colbench_example (2)	0.006	0	33.8	yes	yes	yes	0.7
colbench_text_deepseekr1	0.007	377	18.9				0.0
colbench_text_o4mini	0.009	0	30.9		yes	yes	0.0
colbench_text_gpt41	0.014	0	35.6	yes	yes	yes	0.8
colbench_backend_programming colbench_example (3)	0.015	0	37.2	yes	yes	yes	0.8
colbench_example_agent_gpt41	0.016	0	38.6	yes	yes	yes	0.8
colbench_backend_programming colbench_example (4)	0.024	0	41.2	yes	yes	yes	0.9
colbench_backend_programming colbench_example (5)	0.026	0	37.1				0.0
Hal Generalist Agent (DeepSeek-V3)	0.030	0	34.5				0.0
colbench_backend_programming colbench_example (6)	0.031	0	34.3				0.0
colbench_text_sonnet37	0.034	0	37.4				0.0
colbench_backend_programming colbench_example (7)	0.035	0	40.7		yes	yes	0.3
colbench_backend_programming colbench_example (8)	0.041	0	40.0		yes	yes	0.1
colbench_backend_programming colbench_example (9)	0.053	1	22.3				0.0
Hal Generalist Agent (GPT4.1)	0.056	0	35.9				0.0
Hal Generalist Agent (O4-mini-high)	0.063	0	30.4				0.0
Hal Generalist Agent (DeepSeek-R1)	0.106	0	29.6				0.0
Hal Generalist Agent (O3-low)	0.106	0	25.0				0.0
Hal Generalist Agent (Sonnet3.7)	0.202	0	43.7	yes	yes	yes	0.8

Table S3: ColBench frontend design: 14 systems on 100 tasks, cheapest first. Uncosted: tasks with an outcome but no logged model call, priced at the run's mean costed task. In set (fixed): the frontier set with every cost fixed at its mean. Rival lead: how far the strongest cheaper-or-equal system sits above this one, in paired standard errors (negative when this system is ahead); the simultaneous margin for this board is 2.52.

System	$/task	Uncosted	Acc. %	Frontier	In set	In set (fixed)	Undom. freq.	Best set
colbench_text_deepseekr1	0.009	2	74.9	yes	yes	yes	0.97	
colbench_text_deepseekv3	0.010	16	75.2	yes	yes	yes	0.66	
colbench_text_gemini	0.027	0	81.6	yes	yes	yes	1.00	yes
Hal Generalist Agent (DeepSeek-V3)	0.060	0	71.3				0.00	
colbench_text_o4minilow	0.061	0	81.0		yes	yes	0.25	yes
Hal Generalist Agent (GPT4.1)	0.065	0	72.2				0.00	
Hal Generalist Agent (o4-mini-low)	0.097	0	72.1				0.00	
colbench_text_o4minimedium	0.124	0	82.9	yes	yes	yes	0.90	yes

System	$/task	Uncosted	Acc. %	Frontier	In set	In set (fixed)	Undom. freq.	Best set
Hal Generalist Agent (o3-low)	0.132	0	67.4				0.00	
colbench_text_o4minihigh	0.142	0	82.5		yes	yes	0.35	yes
Hal Generalist Agent (o4-mini-high)	0.144	0	70.7				0.00	
Hal Generalist Agent (DeepSeek-R1)	0.172	0	57.0				0.00	
colbench_text_sonnet	0.713	0	82.8		yes	yes	0.38	yes
Hal Generalist Agent (Sonnet 3.7)	1.048	0	69.3				0.00	

Table S4: CORE-Bench hard: 60 systems on 45 tasks, cheapest first. Uncosted: tasks with an outcome but no logged model call, priced at the run's mean costed task. In set (fixed): the frontier set with every cost fixed at its mean. Rival lead: how far the strongest cheaper-or-equal system sits above this one, in paired standard errors (negative when this system is ahead); the simultaneous margin for this board is 2.97.

System	$/task	Uncosted	Acc. %	Frontier	In set	In set (fixed)	Undom. freq.
HAL Generalist Agent(Deepseek-V3-0324)	0.017	0	8.9	yes	yes	yes	1.00
HAL Generalist Agent (1)	0.046	0	8.9		yes	yes	0.42
HAL Generalist Agent(Deepseek-R1-0528)	0.057	0	8.9		yes	yes	0.26
HAL Generalist Agent (2)	0.062	0	8.9		yes	yes	0.19
Hal Generalist Agent(Gemini-2.0-flash)	0.078	0	4.4		yes	yes	0.01
CORE-Agent (1)	0.093	0	11.1	yes	yes	yes	0.26
CORE-Agent (2)	0.095	1	11.1		yes	yes	0.21
HAL Generalist Agent(o4-mini-low)	0.121	0	15.6	yes	yes	yes	0.35
CORE-Agent(Deepseek-V3.1)	0.136	2	20.0	yes	yes	yes	0.59
CORE-Agent (3)	0.138	0	11.1		yes	yes	0.07
coreagent	0.159	0	20.0		yes	yes	0.41
HAL Generalist Agent (3)	0.190	0	15.6		yes	yes	0.07
HAL Generalist Agent (4)	0.196	0	6.7		yes	yes	0.00
HAL Generalist Agent (5)	0.209	0	15.6		yes	yes	0.04
HAL Generalist Agent (6)	0.210	1	8.9		yes	yes	0.00
CORE-Agent (4)	0.216	5	15.6		yes	yes	0.02
CORE-Agent (5)	0.267	0	17.8		yes	yes	0.07
CORE-Agent (6)	0.281	0	17.8		yes	yes	0.06
HAL Generalist Agent(gpt4.1)	0.296	0	22.2	yes	yes	yes	0.19
Hal Generalist Agent(Deepseek-R1)	0.308	0	2.2				0.00
HAL Generalist Agent(gpt-5-medium)	0.320	1	11.1		yes	yes	0.00
HAL Generalist Agent (7)	0.325	2	13.3		yes	yes	0.00
Core Agent	0.338	0	40.0	yes	yes	yes	0.83
HAL Generalist Agent(o4-mini-high)	0.365	0	35.6		yes	yes	0.47
HAL Generalist Agent (8)	0.395	1	35.6		yes	yes	0.30
HAL Generalist Agent (9)	0.447	11	4.4				0.00
HAL Generalist Agent(o3-medium)	0.470	1	22.2		yes	yes	0.01
HAL Generalist Agent (10)	0.504	0	24.4		yes	yes	0.00
HAL Generalist Agent (11)	0.523	0	6.7				0.00
CORE-Agent (7)	0.527	0	33.3		yes	yes	0.07
CORE-Agent (8)	0.546	0	26.7		yes	yes	0.01
CORE-Agent (9)	0.692	0	24.4		yes	yes	0.00
Core-Agent(Deepseek-R1)	0.788	1	4.4				0.00
HAL Generalist Agent (12)	0.815	0	8.9				0.00
CORE-Agent(Claude-Haiku-4-5)	0.976	0	11.1				0.00
CORE-Agent(Claude-Sonnet-4)	1.140	1	28.9		yes	yes	0.02
HAL Generalist Agent (13)	1.146	0	26.7		yes	yes	0.01
HAL Generalist Agent (14)	1.236	1	31.1		yes	yes	0.03
HAL Generalist Agent(sonnet 3.7)	1.258	0	31.1		yes	yes	0.02
HAL Generalist Agent (15)	1.405	0	22.2		yes	yes	0.00
CORE-Agent (10)	1.450	1	8.9				0.00

System	$/task	Uncosted	Acc. %	Frontier	In set	In set (fixed)	Undom. freq.
HAL Generalist Agent(sonnet 3.7 high)	1.469	0	37.8		yes	yes	0.17
CORE-Agent (11)	1.609	0	24.4		yes	yes	0.00
CORE-Agent (12)	1.622	0	35.6		yes	yes	0.07
CORE-Agent (13)	1.671	0	33.3		yes	yes	0.04
HAL-Generalist-Agent(Claude-Sonnet-4-5)	1.892	0	33.3		yes	yes	0.04
HAL-Generalist-Agent(Claude-Sonnet-4-5-high)	1.950	0	28.9		yes	yes	0.00
CORE-Agent (14)	2.026	0	22.2		yes	yes	0.00
CORE-Agent(Claude-Sonnet-4-5-high)	2.099	1	44.4	yes	yes	yes	0.43
CORE-Agent(Claude- Sonnet-4-5)	2.208	1	37.8		yes	yes	0.11
CORE-Agent(Claude-Sonnet-4-high)	2.232	0	33.3		yes	yes	0.00
Hal Generalist Agent Opus 4.5 high	2.554	1	31.1		yes	yes	0.02
Hal Generalist Agent Opus 4.5	2.831	0	33.3		yes	yes	0.02
Core Agent Opus 4.5 high	3.390	0	42.2		yes	yes	0.17
Core Agent Opus 4.5	3.755	0	42.2		yes	yes	0.12
Coreagent(claude-opus-4-5)	6.082	0	75.6	yes	yes	yes	1.00
HAL Generalist Agent (16)	8.147	1	33.3				0.00
HAL Generalist Agent (17)	8.525	1	35.6				0.00
CORE-Agent (15)	9.164	0	51.1				0.01
CORE-Agent (16)	11.331	0	42.2				0.00

Table S5: GAIA: 36 systems on 165 tasks, cheapest first. Uncosted: tasks with an outcome but no logged model call, priced at the run's mean costed task. In set (fixed): the frontier set with every cost fixed at its mean. Rival lead: how far the strongest cheaper-or-equal system sits above this one, in paired standard errors (negative when this system is ahead); the simultaneous margin for this board is 2.85.

System	$/task	Uncosted	Acc. %	Frontier	In set	In set (fixed)	Undom. fr
HAL Generalist Agent (deepseek-chat-v3-0324)	0.006	0	22.4	yes	yes	yes	1.00
HAL Generalist Agent (gemini-2.0-flash)	0.024	0	32.7	yes	yes	yes	0.98
HAL Generalist Agent (o3-mini-2025-01-31 low)	0.032	0	33.9	yes	yes	yes	0.59
HF Open Deep Research (gemini-2.0-flash)	0.057	0	19.4				0.00
HAL Generalist Agent (o3-mini-2025-01-31 high)	0.086	0	46.7	yes	yes	yes	0.99
HAL Generalist Agent (DeepSeek-V3)	0.089	0	36.4		yes	yes	0.30
HAL Generalist Agent (gpt-4.1-2025-04-14)	0.121	0	49.7	yes	yes	yes	0.63
HAL Generalist Agent (o4-mini-2025-04-16 high)	0.145	0	54.5	yes	yes	yes	0.71
HF Open Deep Research (o3-mini-2025-01-31 hig	0.154	1	18.2				0.00
HAL Generalist Agent (o4-mini-2025-04-16 low)	0.175	0	58.2	yes	yes	yes	0.75
HF Open Deep Research (gpt-4.1-2025-04-14)	0.190	1	50.3		yes	yes	0.06
HF Open Deep Research (o4-mini-2025-04-16 low)	0.195	1	47.9		yes	yes	0.01
HAL Generalist Agent (gpt-5-2025-08-07)	0.219	0	59.4	yes	yes	yes	0.55
HAL Generalist Agent (deepseek-ai/DeepSeek-R1)	0.222	0	30.3				0.00
HF Open Deep Research (deepseek-ai/DeepSeek-V	0.232	0	28.5				0.00
HF Open Deep Research (o3-2025-04-16)	0.283	0	32.7				0.00
HF Open Deep Research (deepseek-ai/DeepSeek-R	0.434	0	24.8				0.00
HF Open Deep Research (o4-mini-2025-04-16 hig	0.482	0	55.8		yes	yes	0.11
HF Open Deep Research (claude-3-7-sonnet-2025 (1)	0.689	0	35.8				0.00
HAL Generalist Agent (claude-3-7-sonnet-20250 (1)	0.742	0	64.2	yes	yes	yes	0.72
HAL Generalist Agent (claude-3-7-sonnet-20250 (2)	0.791	0	56.4		yes	yes	0.04
HAL Generalist Agent (Claude-Haiku-4-5)	0.843	10	56.4		yes	yes	0.05
HAL Generalist Agent (claude-sonnet-4.5) (1)	1.023	0	74.5	yes	yes	yes	0.85
HF Open Deep Research (gpt-5-2025-08-07)	1.024	0	62.4				0.11
HAL Generalist Agent (claude-sonnet-4.5) (2)	1.089	0	70.9		yes	yes	0.25
HAL Generalist Agent (claude-sonnet-4.5) (3)	1.134	0	74.5		yes	yes	0.56
HF Open Deep Research (claude-3-7-sonnet-2025 (2)	2.678	10	37.0				0.00
HAL Generalist Agent (claude-opus-4-20250514) (1)	2.700	64	30.3				0.00

System	$/task	Uncosted	Acc. %	Frontier	In set	In set (fixed)	Undom. fre
HF Open Deep Research (claude-sonnet-4.5) (1)	2.724	0	30.9				0.00
HF Open Deep Research (claude-sonnet-4.5) (2)	3.228	0	30.9				0.00
HAL Generalist Agent (claude-opus-4.1) (1)	3.407	0	68.5		yes	yes	0.00
HAL Generalist Agent (claude-opus-4.1) (2)	3.889	0	64.2				0.00
HAL Generalist Agent (claude-opus-4-20250514) (2)	4.059	1	64.8		yes	yes	0.00
HF Open Deep Research (claude-opus-4.1)	7.920	0	28.5				0.00
HF Open Deep Research (claude-opus-4.1 high)	8.931	0	25.5				0.00
HF Open Deep Research (claude-opus-4)	10.219	0	57.6				0.00

Table S6: Online-Mind2Web†: 32 systems on 300 tasks, cheapest first. Uncosted: tasks with an outcome but no logged model call, priced at the run's mean costed task. In set (fixed): the frontier set with every cost fixed at its mean. Rival lead: how far the strongest cheaper-or-equal system sits above this one, in paired standard errors (negative when this system is ahead); the simultaneous margin for this board is 2.80.

System	$/task	Uncosted	Acc. %	Frontier	In set	In set (fixed)	Undom. freq
SeeAct(gemini-2.0-flash)	0.017	0	26.7	yes	yes	yes	1.00
Browser-Use_test(gemini-2.0-flash)	0.060	0	29.0	yes	yes	yes	0.78
SeeAct(claude-haiku-4-5-20251001)	0.223	0	32.7	yes	yes	yes	0.87
SeeAct(gpt-5-2025-08-07_minimal_reasoning_eff	0.263	0	30.3		yes	yes	0.17
SeeAct(claude-haiku-4-5-20251001_thinking_hig	0.297	0	36.3	yes	yes	yes	0.92
SeeAct(gemini-2.5-pro-preview-03-25)	0.512	0	39.7	yes	yes	yes	0.85
SeeAct(o4-mini-2025-04-16_low_reasoning_effor	0.541	0	31.7		yes	yes	0.00
SeeAct(gpt-5-2025-08-07_medium_reasoning_effo	0.570	0	42.3	yes	yes	yes	0.82
Browser-Use(o4-mini-2025-04-16_low_reasoning_	0.671	0	18.3				0.00
SeeAct(claude-sonnet-4-5-20250929)	0.683	0	35.0		yes	yes	0.00
Browser-Use_test(DeepSeek-V3)	0.716	0	32.3				0.00
SeeAct(o4-mini-2025-04-16_high_reasoning_effo	0.763	0	32.0				0.00
Browser-Use(gpt-4.1-2025-04-14)	0.789	0	36.3		yes	yes	0.03
SeeAct(claude-sonnet-4-20250514)	0.821	0	36.7		yes	yes	0.01
SeeAct(o3-2025-04-16)	0.862	0	39.0		yes	yes	0.08
SeeAct(gpt-4.1-2025-04-14)	0.904	0	30.3				0.00
Browser-Use_test(DeepSeek-R1)	0.936	0	25.3				0.00
SeeAct(claude-3-7-sonnet-20250219)	0.973	0	28.3				0.00
SeeAct(claude-sonnet-4-5-20250929_thinking_hi	0.977	0	38.0		yes	yes	0.03
Browser-Use(o4-mini-2025-04-16_high_reasoning	0.993	0	20.0				0.00
SeeAct(claude-sonnet-4-20250514_thinking_high	1.088	0	36.7		yes	yes	0.01
Browser-Use_test(gemini-2.5-pro-preview-03-25)	1.105	0	41.0		yes	yes	0.26
SeeAct(gpt-5-2025-08-07_high_reasoning_effort)	1.143	0	41.7		yes	yes	0.22
Browser-Use(gpt-5-2025-08-07_minimal_reasonin	1.157	0	22.0				0.00
SeeAct(claude-3-7-sonnet-20250219_thinking_hi	1.225	0	30.3				0.00
Browser-Use(o3-2025-04-16)	1.239	0	29.0				0.00
Browser-Use(gpt-5-2025-08-07_medium_reasoning	2.454	0	32.0				0.00
Browser-Use_test(claude-3-7-sonnet-20250219)	3.088	0	38.3		yes	yes	0.03
Browser-Use(gpt-5-2025-08-07_high_reasoning_e	3.565	0	35.7		yes	yes	0.00
Browser-Use_test(claude-3-7-sonnet-20250219_t	3.840	0	39.3		yes	yes	0.06
Browser-Use_test(claude-sonnet-4-20250514)	5.258	0	40.0		yes	yes	0.07
Browser-Use_test(claude-sonnet-4-20250514_thi	5.366	0	39.3		yes	yes	0.03

Table S7: SciCode: 37 systems on 65 tasks, cheapest first. Uncosted: tasks with an outcome but no logged model call, priced at the run's mean costed task. In set (fixed): the frontier set with every cost fixed at its mean. Rival lead: how far the strongest cheaper-or-equal system sits above this one, in paired standard errors (negative when this system is ahead); the simultaneous margin for this board is 3.07.

System	$/task	Uncosted	Acc. %	Frontier	In set	In set (fixed)	Undom. fre
Scicode Zero Shot Agent (gemini-2.0-flash)	0.002	0	1.5	yes	yes	yes	1.00
Scicode Zero Shot Agent (deepseek-ai/DeepSeek (1)	0.012	0	3.1	yes	yes	yes	0.63
Scicode Zero Shot Agent (o4-mini-2025-04-16 l	0.026	0	9.2	yes	yes	yes	0.98
Scicode Zero Shot Agent (gpt-4.1-2025-04-14)	0.039	0	6.2		yes	yes	0.19
Scicode Tool Calling Agent (gemini-2.0-flash)	0.041	0	1.5		yes	yes	0.00
Scicode Zero Shot Agent (o4-mini-2025-04-16)	0.042	0	6.2		yes	yes	0.07
Scicode Zero Shot Agent (claude-3-7-sonnet-20 (1)	0.077	0	3.1		yes	yes	0.00
Scicode Zero Shot Agent (claude-3-7-sonnet-20 (2)	0.078	0	0.0		yes	yes	0.00
Scicode Zero Shot Agent (o4-mini-2025-04-16 h	0.083	1	6.2		yes	yes	0.04
Scicode Zero Shot Agent (o3-2025-04-16)	0.089	0	4.6		yes	yes	0.01
Scicode Zero Shot Agent (deepseek-ai/DeepSeek (2)	0.143	1	0.0		yes	yes	0.00
Scicode Tool Calling Agent (o4-mini-2025-04-1 (1)	0.320	0	4.6		yes	yes	0.00
Hal Generalist Agent (o3-2025-04-16) (1)	0.355	0	6.2		yes	yes	0.06
Hal Generalist Agent (gpt-4.1-2025-04-14)	0.359	0	1.5		yes	yes	0.00
Hal Generalist Agent (o3-2025-04-16) (2)	0.382	0	3.1		yes	yes	0.00
Scicode Tool Calling Agent (gpt-4.1-2025-04-1	0.394	0	1.5		yes	yes	0.00
Scicode Tool Calling Agent (o4-mini-2025-04-1 (2)	0.399	0	3.1		yes	yes	0.00
Scicode Tool Calling Agent (deepseek-ai/DeepS (1)	0.404	0	0.0		yes	yes	0.00
Scicode Tool Calling Agent (o4-mini-2025-04-1 (3)	0.494	0	4.6		yes	yes	0.00
Hal Generalist Agent (o4-mini-2025-04-16)	0.576	0	1.5		yes	yes	0.00
Hal Generalist Agent (o4-mini-2025-04-16 high)	0.623	1	1.5		yes	yes	0.00
Scicode Tool Calling Agent (o3-2025-04-16)	0.722	0	9.2		yes	yes	0.26
HAL Generalist Agent (claude-3-7-sonnet-20250 (1)	0.990	4	3.1		yes	yes	0.00
Hal Generalist Agent (o4-mini-2025-04-16 low)	1.029	0	6.2		yes	yes	0.01
Scicode Tool Calling Agent (gpt-5)	1.442	0	6.2		yes	yes	0.01
Scicode Tool Calling Agent (claude-3-7-sonnet (1)	1.499	0	3.1		yes	yes	0.00
Scicode Tool Calling Agent (claude-3-7-sonnet (2)	1.599	0	4.6		yes	yes	0.01
Scicode Tool Calling Agent (claude-sonnet-4-5	1.604	0	3.1		yes	yes	0.00
HAL Generalist Agent (deepseek-ai/DeepSeek-V3)	1.714	1	0.0		yes	yes	0.00
Scicode Tool Calling Agent (claude-haiku-4-5)	1.807	0	0.0		yes	yes	0.00
Scicode Tool Calling Agent (claude-sonnet-4.5	1.818	0	1.5		yes	yes	0.00
Scicode Tool Calling Agent (deepseek-ai/DeepS (2)	2.142	0	0.0		yes	yes	0.00
HAL Generalist Agent (claude-3-7-sonnet-20250 (2)	2.895	0	3.1		yes	yes	0.01
HAL Generalist Agent (deepseek-ai/DeepSeek-R1)	3.744	0	0.0		yes	yes	0.00
SciCode Tool Calling Agent (claude-opus-4.1-2	7.811	0	7.7		yes	yes	0.06
Scicode Tool Calling Agent (claude-opus-4.1 h	9.128	0	6.2		yes	yes	0.00
Scicode Tool Calling Agent (claude-opus-4.1)	9.617	0	7.7		yes	yes	0.04

Table S8: ScienceAgentBench: 25 systems on 102 tasks, cheapest first. Uncosted: tasks with an outcome but no logged model call, priced at the run's mean costed task. In set (fixed): the frontier set with every cost fixed at its mean. Rival lead: how far the strongest cheaper-or-equal system sits above this one, in paired standard errors (negative when this system is ahead); the simultaneous margin for this board is 2.76.

System	$/task	Uncosted	Acc. %	Frontier	In set	In set (fixed)	Undom. fre
SAB Self-Debug gemini-2-0-flash	0.002	0	12.7	yes	yes	yes	1.00
SAB Self-Debug DS-V3	0.010	0	15.7	yes	yes	yes	0.80
SAB Self-Debug o4-mini low	0.019	0	27.5	yes	yes	yes	0.99

System	$/task	Uncosted	Acc. %	Frontier	In set	In set (fixed)	Undom. fre
SAB Self-Debug Claude-Haiku-4-5	0.026	0	18.6		yes	yes	0.04
SAB Self-Debug Claude-Haiku-4-5 High	0.033	0	23.5		yes	yes	0.15
sab_selfdebug_gpt_41	0.034	0	24.5		yes	yes	0.20
SAB Self-Debug o4-mini high	0.055	0	27.5		yes	yes	0.27
SAB Self-Debug o3 medium	0.057	0	33.3	yes	yes	yes	0.77
SAB Self-Debug Claude-3-7 low	0.064	0	25.5		yes	yes	0.04
SAB Self-Debug Claude-3-7	0.070	0	22.5		yes	yes	0.01
SAB Self-Debug Claude-Sonnet-4-5	0.072	0	29.4		yes	yes	0.11
SAB Self-Debug Claude-Sonnet-4-5 High	0.073	0	30.4		yes	yes	0.18
SAB Self-Debug GPT-5-Medium	0.089	0	30.4		yes	yes	0.08
sab_selfdebug_dsr1	0.089	0	23.5		yes	yes	0.00
SAB Self-Debug gemini-2-5-pro	0.110	0	18.6				0.00
SAB Self-Debug Claude-3-7 high	0.115	0	30.4		yes	yes	0.09
HAL Generalist Agent (o3-2025-04-16)	0.138	0	9.8				0.00
HAL Generalist Agent (gpt-4.1-2025-04-14)	0.232	0	6.9				0.00
HAL Generalist Agent (deepseek-ai/DeepSeek-V3)	0.273	0	1.0				0.00
SAB Self-Debug Claude-Opus-4-1	0.327	0	27.5		yes	yes	0.03
SAB Self-Debug Claude-Opus-4-1-High	0.331	0	26.5		yes	yes	0.02
HAL Generalist Agent (o4-mini-2025-04-16 high)	0.331	0	21.6		yes	yes	0.00
HAL Generalist Agent (o4-mini-2025-04-16 low)	0.336	0	19.6				0.00
HAL Generalist Agent (claude-3-7-sonnet-20250 (1)	0.404	0	10.8				0.00
HAL Generalist Agent (claude-3-7-sonnet-20250 (2)	0.473	0	17.6				0.00

Table S9: SWE-bench Verified mini: 39 systems on 50 tasks, cheapest first. Uncosted: tasks with an outcome but no logged model call, priced at the run's mean costed task. In set (fixed): the frontier set with every cost fixed at its mean. Rival lead: how far the strongest cheaper-or-equal system sits above this one, in paired standard errors (negative when this system is ahead); the simultaneous margin for this board is 3.00.

System	$/task	Uncosted	Acc. %	Frontier	In set	In set (fixed)	Undom. freq.
hal_generalist_agent_gemini20flash	0.073	0	2.0	yes	yes	yes	0.63
hal_generalist_agent_o3mini20250131_high	0.075	0	0.0		yes	yes	0.20
hal_generalist_agent_o3mini20250131_low	0.077	0	0.0		yes	yes	0.40
My Agent(gemini/gemini-2.0-flash)	0.094	0	22.0	yes	yes	yes	1.00
HAL Generalist Agent (o4-mini-2025-04-16 high)	0.229	0	2.0				0.00
hal_generalist_agent_deepseekaideepseekv3	0.302	0	10.0		yes	yes	0.02
HAL Generalist Agent (gpt-4.1-2025-04-14)	0.324	0	2.0				0.00
HAL Generalist Agent (o4-mini-2025-04-16 low)	0.398	0	6.0		yes	yes	0.01
HAL Generalist (gpt-5-2025-08-07)	0.451	0	14.0		yes	yes	0.09
SWE-Agent(claude-3-7-sonnet-20250219) (1)	0.667	0	54.0	yes	yes	yes	0.72
SWE-Agent(claude-3-7-sonnet-20250219) (2)	0.681	0	52.0		yes	yes	0.53
My Agent(claude-3-7-sonnet-20250219)	0.794	0	50.0		yes	yes	0.11
SWE-Agent(gpt-5-2025-08-07)	1.020	0	64.0	yes	yes	yes	0.89
My Agent(o3-mini)	1.212	1	46.0		yes	yes	0.02
SWE-Agent(gpt-4.1)	1.265	0	46.0		yes	yes	0.00
HAL Generalist (claude-haiku-4-5-20251001) (1)	1.306	0	44.0		yes	yes	0.00
hal_generalist_agent_deepseekaideepseekr1	1.467	0	6.0				0.00
HAL Generalist Agent (gpt-4o-2024-11-20)	1.496	0	2.0				0.00
SWE-Agent(claude-sonnet-4-5-20250929) (1)	1.509	0	68.0	yes	yes	yes	0.59
HAL Generalist (claude-3-7-sonnet-20250219)	1.536	0	38.0				0.00
SWE-Agent(claude-sonnet-4-5-20250929) (2)	1.573	0	72.0	yes	yes	yes	0.75
HAL Generalist (claude-haiku-4-5-20251001) (2)	1.744	0	24.0				0.00
HAL Generalist (o3-2025-04-16)	1.884	0	0.0				0.00
HAL Generalist (claude-sonnet-4-5-20250929) (1)	1.999	2	40.0				0.00
My Agent(claude-opus-4-20250514)	2.119	0	60.0		yes	yes	0.01

System	$/task	Uncosted	Acc. %	Frontier	In set	In set (fixed)	Undom. freq.
HAL Generalist Agent (claude-3-7-sonnet-20250	2.140	1	26.0				0.00
SWE-Agent(o4-mini-2025-04-16) (1)	2.318	1	52.0				0.00
SWE-Agent(o4-mini-2025-04-16) (2)	2.445	2	56.0		yes	yes	0.00
HAL Generalist (claude-sonnet-4-5-20250929) (2)	2.616	1	34.0				0.00
SWE-Agent(claude-opus-4-1-20250805) (1)	2.623	0	66.0		yes	yes	0.10
SWE-Agent(claude-opus-4-1-20250805) (2)	2.811	0	54.0		yes	yes	0.00
SWE-Agent(claude-opus-4-1-20250805) (3)	3.403	0	68.0		yes	yes	0.13
My Agent(gpt-4o)	3.507	0	28.0				0.00
HAL Generalist (claude-haiku-4-5)	4.171	0	24.0				0.00
My Agent(o1)	4.478	0	40.0				0.00
HAL Generalist (claude-opus-4-20250514) (1)	7.648	0	34.0				0.00
HAL Generalist (claude-opus-4-1-20250805) (1)	7.999	0	56.0		yes	yes	0.00
HAL Generalist (claude-opus-4-20250514) (2)	8.068	0	30.0				0.00
HAL Generalist (claude-opus-4-1-20250805) (2)	9.553	0	52.0		yes	yes	0.00

Table S10: tau-bench airline: 47 systems on 50 tasks, cheapest first. Uncosted: tasks with an outcome but no logged model call, priced at the run's mean costed task. In set (fixed): the frontier set with every cost fixed at its mean. Rival lead: how far the strongest cheaper-or-equal system sits above this one, in paired standard errors (negative when this system is ahead); the simultaneous margin for this board is 2.81.

System	$/task	Uncosted	Acc. %	Frontier	In set	In set (fixed)	Undom. fr
HAL Generalist Agent (gemini-2.0-flash)	0.020	0	22.0	yes	yes	yes	1.00
Taubench ToolCalling (gemini-2.0-flash-001)	0.028	0	28.0	yes	yes	yes	0.71
TAU-bench FewShot (gemini-2.0-flash)	0.044	0	44.0	yes	yes	yes	0.86
Taubench ToolCalling (o4-mini-2025-04-16 low)	0.045	0	36.0		yes	yes	0.40
Taubench ToolCalling (gpt-4.1-2025-04-14)	0.077	0	36.0		yes	yes	0.09
Taubench ToolCalling (o4-mini-2025-04-16 high)	0.084	0	56.0	yes	yes	yes	0.78
Taubench ToolCalling (o3-2025-04-16)	0.090	0	54.0		yes	yes	0.45
HAL Generalist Agent (deepseek-ai/DeepSeek-V3)	0.107	0	18.0				0.00
HAL Generalist Agent (gpt-4.1-2025-04-14)	0.115	0	16.0				0.00
TAU-bench FewShot (o3-mini-2025-01-31 low)	0.128	0	40.0		yes	yes	0.00
Taubench ToolCalling (deepseek-v3)	0.130	1	44.0		yes	yes	0.02
TAU-bench FewShot (o3-mini-2025-01-31)	0.140	0	48.0		yes	yes	0.05
TAU-bench FewShot (o4-mini-2025-04-16 high)	0.145	0	60.0	yes	yes	yes	0.52
TAU-bench FewShot (o4-mini-2025-04-16 low)	0.149	0	48.0		yes	yes	0.03
HAL Generalist Agent (o4-mini-2025-04-16 low)	0.173	0	22.0				0.00
HAL Generalist Agent (o4-mini-2025-04-16 high)	0.174	0	18.0				0.00
TAU-bench FewShot (o3-2025-04-03)	0.178	0	44.0		yes	yes	0.00
Taubench ToolCalling (gpt-5-2025-08-07)	0.182	0	48.0		yes	yes	0.02
TAU-bench FewShot (gpt-4.1-2025-04-14)	0.213	0	56.0		yes	yes	0.17
TAU-bench FewShot (o3-2025-04-16 medium)	0.242	0	48.0		yes	yes	0.00
taubench_fewshot_o3mini20250131_high	0.243	0	40.0		yes	yes	0.00
TauBench Few-Shot Minimal Reasoning (gpt-5)	0.248	0	50.0		yes	yes	0.03
TauBench Few-Shot High Reasoning (gpt-5)	0.259	0	52.0		yes	yes	0.02
HAL Generalist Agent (o3-2025-04-16 medium)	0.263	0	20.0				0.00
Taubench ToolCalling (deepseek-r1)	0.283	0	36.0				0.00
HAL Generalist Agent (deepseek-ai/DeepSeek-R1)	0.302	0	10.0				0.00
TAU-bench FewShot (deepseek-ai/DeepSeek-V3)	0.306	0	34.0				0.00
TAU-bench FewShot (deepseek-ai/DeepSeek-R1)	0.317	0	36.0				0.00
Taubench ToolCalling (claude-3.7-sonnet)	0.324	0	44.0		yes	yes	0.00
Taubench ToolCalling (claude-3.7-sonnet high)	0.338	0	52.0		yes	yes	0.04
HAL Generalist High Reasoning (gpt-5)	0.447	0	30.0				0.00
TAU-bench FewShot (gpt-4o-2024-11-20)	0.465	0	40.0		yes	yes	0.00
HAL Generalist Minimal Reasoning (gpt-5)	0.517	0	38.0		yes	yes	0.00

System	$/task	Uncosted	Acc. %	Frontier	In set	In set (fixed)	Undom. fr
HAL Generalist Agent (claude-3-7-sonnet-20250 (1)	0.677	0	44.0		yes	yes	0.01
TAU-bench FewShot (claude-3-7-sonnet-20250219) (1)	0.712	0	34.0				0.00
TAU-bench FewShot (claude-3-7-sonnet-20250219) (2)	0.726	0	60.0		yes	yes	0.27
HAL Generalist Agent (claude-3-7-sonnet-20250 (2)	0.824	0	56.0		yes	yes	0.10
Taubench ToolCalling (claude-opus-4.1)	1.415	0	50.0		yes	yes	0.01
Taubench ToolCalling (claude-opus-4.1 high)	1.518	0	52.0		yes	yes	0.01
HAL Generalist High Reasoning (claude-opus-4.	2.793	0	32.0				0.00
TauBench FewShot No Reasoning (claude-opus-4.	2.942	0	54.0		yes	yes	0.01
TauBench FewShot High Reasoning (claude-opus-	2.986	0	62.0	yes	yes	yes	0.22
HAL Generalist (claude-opus-4-20250514)	2.987	0	44.0		yes	yes	0.00
HAL Generalist High Reasoning (claude-opus-4-	2.992	0	44.0		yes	yes	0.00
TauBench Few Shot High Reasoning (claude-opus	3.138	0	66.0	yes	yes	yes	0.44
HAL Generalist No Reasoning (claude-opus-4.1)	3.593	0	54.0		yes	yes	0.01
TauBench Few Shot (claude-opus-4-20250514)	3.633	0	56.0		yes	yes	0.01

Table S11: USACO: 12 systems on 307 tasks, cheapest first. Uncosted: tasks with an outcome but no logged model call, priced at the run's mean costed task. In set (fixed): the frontier set with every cost fixed at its mean. Rival lead: how far the strongest cheaper-or-equal system sits above this one, in paired standard errors (negative when this system is ahead); the simultaneous margin for this board is 2.51.

System	$/task	Uncosted	Acc. %	Frontier	In set	In set (fixed)	Undom. f
usaco_episodic (1)	0.020	0	39.1	yes	yes	yes	1.00
USACO Episodic + Semantic (o4-mini-2025-04-16 (1)	0.034	0	30.9				0.00
usaco_episodic (2)	0.037	0	35.2		yes	yes	0.05
usaco_episodic (3)	0.046	0	45.0	yes	yes	yes	1.00
USACO Episodic + Semantic (o4-mini-2025-04-16 (2)	0.076	18	58.0	yes	yes	yes	1.00
USACO Episodic + Semantic (o3-2025-04-16)	0.098	14	46.3				0.00
usaco_episodic (4)	0.106	16	49.8				0.00
USACO Episodic + Semantic (gpt-5-2025-08-07)	0.113	24	69.7	yes	yes	yes	1.00
USACO Episodic + Semantic (claude-3-7-sonnet-	0.126	0	29.3				0.00
usaco_episodic (5)	0.154	47	38.1				0.00
USACO Episodic + Semantic (claude-opus-4.1-20 (1)	0.436	0	51.5				0.00
USACO Episodic + Semantic (claude-opus-4.1-20 (2)	0.450	0	48.2				0.00

S2. Certification of frontier entrants

The steps behind Table 4 of the main text.

Table S12: AssistantBench: frontier entrants in arrival order. Holder: the most accurate cheaper-or-equal earlier system. p raw: selective record test; g: submissions on the entrant's date; $p = \min(1, g \cdot p_{\text{raw}})$. Levels under alpha-spending and the online fallback, with the certification decision.

t	Date	Entrant	Holder	Margin	p raw	g	p	Spending
1	2025-08-11	HAL Generalist Agent (claude-opus	HAL Generalist Agent (deepsee	+3.0	1.000	1	1.000	0.0304

Table S13: ColBench backend programming: frontier entrants in arrival order. Holder: the most accurate cheaper-or-equal earlier system. p raw: selective record test; g: submissions on the entrant's date; $p = \min(1, g \cdot p_{\text{raw}})$. Levels under alpha-spending and the online fallback, with the certification decision.

t	Date	Entrant	Holder	Margin	p raw	g	p	Spending
1	2025-04-20	colbench_text_gpt41	colbench_text_gemmini2flash	+2.7	0.088	1	0.088	0.0304
2	2025-05-23	Hal Generalist Agent (Sonnet3.7)	colbench_text_sonnet37	+6.3	0.004	1	0.004	0.0076
3	2025-09-22	colbench_backend_programming colb	colbench_text_gpt41	+5.6	0.000	9	0.002	0.0034
3	2025-09-22	colbench_example_agent_gpt41	colbench_text_gpt41	+3.0	0.052	9	0.469	0.0034
3	2025-09-22	colbench_backend_programming colb	colbench_text_gpt41	+1.6	0.367	9	1.000	0.0034
3	2025-09-22	colbench_backend_programming colb	colbench_text_gemmini2flash	+0.9	0.548	9	1.000	0.0034
4	2025-09-23	colbench_backend_programming colb	colbench_text_deepseekv3	+3.5	0.004	1	0.004	0.0019

Table S14: ColBench frontend design: frontier entrants in arrival order. Holder: the most accurate cheaper-or-equal earlier system. p raw: selective record test; g: submissions on the entrant's date; $p = \min(1, g \cdot p_{\text{raw}})$. Levels under alpha-spending and the online fallback, with the certification decision.

t	Date	Entrant	Holder	Margin	p raw	g	p	Spending	Cert.	Fallba
1	2025-04-23	colbench_text_deepseekv3	colbench_text_deepseekr1	+0.3	0.755	1	0.755	0.0304		0.03

Table S15: CORE-Bench hard: frontier entrants in arrival order. Holder: the most accurate cheaper-or-equal earlier system. p raw: selective record test; g: submissions on the entrant's date; $p = \min(1, g \cdot p_{\text{raw}})$. Levels under alpha-spending and the online fallback, with the certification decision.

t	Date	Entrant	Holder	Margin	p raw	g	p	Spendi
1	2025-04-19	CORE-Agent	coreagent	+15.6	0.092	8	0.738	0.03
1	2025-04-19	CORE-Agent	coreagent	+13.3	0.145	8	1.000	0.03
2	2025-05-15	HAL Generalist Agent	coreagent	+4.4	1.000	4	1.000	0.00
3	2025-08-06	CORE-Agent	CORE-Agent	+15.6	0.092	4	0.369	0.00
4	2025-08-19	HAL Generalist Agent(o4-mini-high)	coreagent	+15.6	0.065	4	0.262	0.00
4	2025-08-19	HAL Generalist Agent(gpt4.1)	coreagent	+2.2	1.000	4	1.000	0.00
4	2025-08-19	HAL Generalist Agent(o4-mini-low)	CORE-Agent	+4.4	1.000	4	1.000	0.00
5	2025-08-20	HAL Generalist Agent(sonnet 3.7 h	HAL Generalist Agent(o4-mini-	+2.2	1.000	4	1.000	0.00
6	2025-08-21	CORE-Agent(Deepseek-V3.1)	HAL Generalist Agent(o4-mini-	+4.4	1.000	3	1.000	0.00
7	2025-10-01	CORE-Agent(Claude-Sonnet-4-5-high)	HAL Generalist Agent(sonnet 3	+6.7	0.607	2	1.000	0.00
8	2025-11-22	Core Agent	HAL Generalist Agent(gpt4.1)	+17.8	0.118	2	0.236	0.00
9	2025-12-15	Coreagent(claude-opus-4-5)	CORE-Agent(Claude-Sonnet-4-5-	+31.1	0.002	1	0.002	0.00

Table S16: GAIA: frontier entrants in arrival order. Holder: the most accurate cheaper-or-equal earlier system. p raw: selective record test; g: submissions on the entrant's date; $p = \min(1, g \cdot p_{\text{raw}})$. Levels under alpha-spending and the online fallback, with the certification decision.

t	Date	Entrant	Holder	Margin	p raw	g	p	Spendin
1	2025-04-17	HF Open Deep Research (gpt-4.1-20	HAL Generalist Agent (gpt-4.1	+0.6	1.000	5	1.000	0.030
2	2025-04-18	HF Open Deep Research (o4-mini-20	HF Open Deep Research (gpt-4.	+5.5	0.314	1	0.314	0.007
3	2025-04-20	HAL Generalist Agent (o4-mini-202	HAL Generalist Agent (gpt-4.1	+4.8	0.438	1	0.438	0.003

t	Date	Entrant	Holder	Margin	p raw	g	p	Spendin
4	2025-04-23	HAL Generalist Agent (o4-mini-202	HAL Generalist Agent (o4-mini	+3.6	0.479	1	0.479	0.001
5	2025-04-25	HAL Generalist Agent (claude-3-7-	HAL Generalist Agent (o4-mini	+6.1	0.265	2	0.529	0.001
6	2025-08-05	HAL Generalist Agent (claude-opus	HAL Generalist Agent (claude-	+0.6	1.000	2	1.000	0.000
7	2025-09-24	HAL Generalist Agent (claude-opus	HAL Generalist Agent (claude-	+4.2	0.419	1	0.419	0.000
8	2025-09-26	HAL Generalist Agent (gpt-5-2025-	HAL Generalist Agent (o4-mini	+1.2	1.000	1	1.000	0.000
9	2025-09-30	HAL Generalist Agent (claude-sonn	HAL Generalist Agent (claude-	+6.7	0.185	1	0.185	0.000
10	2025-10-01	HAL Generalist Agent (claude-sonn	HAL Generalist Agent (claude-	+3.6	0.406	3	1.000	0.000
11	2025-10-14	HAL Generalist Agent (claude-sonn	HAL Generalist Agent (claude-	+10.3	0.024	2	0.048	0.000

Table S17: Online-Mind2Web: frontier entrants in arrival order. Holder: the most accurate cheaper-or-equal earlier system. p raw: selective record test; g: submissions on the entrant's date; $p = \min(1, g \cdot p_{\text{raw}})$. Levels under alpha-spending and the online fallback, with the certification decision.

t	Date	Entrant	Holder	Margin	p raw	g	p	Spendin
1	2025-05-05	Browser-Use_test(claude-3-7-sonne	Browser-Use(gpt-4.1-2025-04-1	+2.0	0.628	2	1.000	0.030
2	2025-05-09	Browser-Use_test(DeepSeek-V3)	Browser-Use(o4-mini-2025-04-1	+14.0	0.000	2	0.000	0.007
3	2025-05-18	SeeAct(o3-2025-04-16)	Browser-Use(gpt-4.1-2025-04-1	+2.7	0.517	1	0.517	0.003
4	2025-08-10	Browser-Use_test(claude-sonnet-4-	SeeAct(gemini-2.5-pro-preview	+0.3	1.000	4	1.000	0.001
4	2025-08-10	Browser-Use_test(gemini-2.0-flash)	SeeAct(gemini-2.0-flash)	+2.3	0.464	4	1.000	0.001
5	2025-08-11	Browser-Use_test(gemini-2.5-pro-p	SeeAct(gemini-2.5-pro-preview	+1.3	0.834	2	1.000	0.001
5	2025-08-11	SeeAct(gpt-5-2025-08-07_minimal_r	Browser-Use_test(gemini-2.0-f	+1.3	0.810	2	1.000	0.001
6	2025-08-13	SeeAct(gpt-5-2025-08-07_medium_re	SeeAct(gemini-2.5-pro-preview	+2.7	0.438	2	0.877	0.000
7	2025-10-28	SeeAct(claude-haiku-4-5-20251001_	SeeAct(gpt-5-2025-08-07_minim	+6.0	0.039	4	0.158	0.000
7	2025-10-28	SeeAct(claude-haiku-4-5-20251001)	Browser-Use_test(gemini-2.0-f	+3.7	0.272	4	1.000	0.000

Table S18: ScienceAgentBench: frontier entrants in arrival order. Holder: the most accurate cheaper-or-equal earlier system. p raw: selective record test; g: submissions on the entrant's date; $p = \min(1, g \cdot p_{\text{raw}})$. Levels under alpha-spending and the online fallback, with the certification decision.

t	Date	Entrant	Holder	Margin	p raw	g	p	Spending	Ce
1	2025-04-17	SAB Self-Debug Claude-3-7 high	SAB Self-Debug Claude-3-7 low	+4.9	0.383	2	0.767	0.0304	
2	2025-04-18	sab_selfdebug_gpt_41	SAB Self-Debug DS-V3	+8.8	0.049	2	0.098	0.0076	
3	2025-04-21	SAB Self-Debug o3 medium	sab_selfdebug_gpt_41	+8.8	0.136	3	0.408	0.0034	
3	2025-04-21	SAB Self-Debug o4-mini high	sab_selfdebug_gpt_41	+2.9	0.678	3	1.000	0.0034	
3	2025-04-21	SAB Self-Debug o4-mini low	SAB Self-Debug DS-V3	+11.8	0.034	3	0.103	0.0034	

Table S19: SWE-bench Verified mini: frontier entrants in arrival order. Holder: the most accurate cheaper-or-equal earlier system. p raw: selective record test; g: submissions on the entrant's date; $p = \min(1, g \cdot p_{\text{raw}})$. Levels under alpha-spending and the online fallback, with the certification decision.

t	Date	Entrant	Holder	Margin	p raw	g	p	Spending
1	2025-04-06	My Agent(o1)	My Agent(gpt-4o)	+12.0	0.227	1	0.227	0.0304
2	2025-08-08	SWE-Agent(claude-opus-4-1-2025080	My Agent(claude-3-7-sonnet-20	+4.0	1.000	1	1.000	0.0076
3	2025-10-31	SWE-Agent(claude-opus-4-1-2025080	SWE-Agent(claude-opus-4-1-202	+14.0	0.065	9	0.589	0.0034
3	2025-10-31	SWE-Agent(claude-opus-4-1-2025080	My Agent(claude-3-7-sonnet-20	+16.0	0.028	9	0.256	0.0034
3	2025-10-31	SWE-Agent(gpt-5-2025-08-07)	My Agent(claude-3-7-sonnet-20	+14.0	0.065	9	0.589	0.0034

t	Date	Entrant	Holder	Margin	p raw	g	p	Spending
3	2025-10-31	SWE-Agent(claude-3-7-sonnet-20250	hal_generalist_agent_deepseek	+44.0	0.000	9	0.000	0.0034
4	2025-11-01	SWE-Agent(claude-sonnet-4-5-20250	SWE-Agent(gpt-5-2025-08-07)	+8.0	0.501	1	0.501	0.0019
5	2025-11-02	SWE-Agent(claude-sonnet-4-5-20250	SWE-Agent(gpt-5-2025-08-07)	+4.0	1.000	1	1.000	0.0012
6	2025-11-09	HAL Generalist (gpt-5-2025-08-07)	hal_generalist_agent_deepseek	+4.0	1.000	4	1.000	0.0008
7	2025-11-10	My Agent(gemini/gemini-2.0-flash)	hal_generalist_agent_gemini20	+20.0	0.008	1	0.008	0.0006

Table S20: tau-bench airline: frontier entrants in arrival order. Holder: the most accurate cheaper-or-equal earlier system. p raw: selective record test; g: submissions on the entrant's date; $p = \min(1, g \cdot p_{\text{raw}})$. Levels under alpha-spending and the online fallback, with the certification decision.

t	Date	Entrant	Holder	Margin	p raw	g	p	Spendin
1	2025-04-07	TAU-bench FewShot (claude-3-7-son	TAU-bench FewShot (o3-mini-20	+12.0	0.319	2	0.639	0.030
2	2025-04-20	TAU-bench FewShot (o4-mini-2025-0	TAU-bench FewShot (o3-mini-20	+12.0	0.227	5	1.000	0.007
3	2025-06-25	TAU-bench FewShot (gemini-2.0-fla	HAL Generalist Agent (gemini-	+22.0	0.035	1	0.035	0.003
4	2025-08-01	TauBench Few Shot High Reasoning	TAU-bench FewShot (claude-3-7	+6.0	0.629	2	1.000	0.001
5	2025-08-05	TauBench FewShot High Reasoning (	TAU-bench FewShot (claude-3-7	+2.0	1.000	4	1.000	0.001
6	2025-10-13	Taubench ToolCalling (gemini-2.0-	HAL Generalist Agent (gemini-	+6.0	0.664	6	1.000	0.000
7	2025-10-20	Taubench ToolCalling (o4-mini-202	TAU-bench FewShot (gemini-2.0	+12.0	0.344	6	1.000	0.000

Table S21: USACO: frontier entrants in arrival order. Holder: the most accurate cheaper-or-equal earlier system. p raw: selective record test; g: submissions on the entrant's date; $p = \min(1, g \cdot p_{\text{raw}})$. Levels under alpha-spending and the online fallback, with the certification decision.

t	Date	Entrant	Holder	Margin	p raw	g	p	Spend
1	2025-04-14	usaco_episodic	usaco_episodic	+5.9	0.011	3	0.033	0.03
2	2025-04-15	usaco_episodic	usaco_episodic	+4.9	0.053	1	0.053	0.00
3	2025-04-29	USACO Episodic + Semantic (o3-202	usaco_episodic	+1.3	0.789	1	0.789	0.00
4	2025-05-05	USACO Episodic + Semantic (o4-min	usaco_episodic	+13.0	0.000	1	0.000	0.00
5	2025-08-25	USACO Episodic + Semantic (gpt-5-	USACO Episodic + Semantic (o4	+11.7	0.000	3	0.000	0.00

S3. The frontier winner's curse, system by system

Table S22: Selection-adjusted accuracy of every system on an observed frontier. Pool: the number of systems that cost no more (the system included); a pool of one has no selection and no correction.

Benchmark	System	Pool	Reported	Iterated bootstrap [95% CI]	Con
AssistantBench	HAL Generalist Agent (gemini-2.0-fl	1	3.0	3.0 [-2.9, 9.0]	3.0 [
	HAL Generalist Agent (deepseek-ai/D	5	12.1	10.9 [-0.0, 21.2]	11.6
	HAL Generalist Agent (claude-opus-4	31	15.2	9.8 [-0.6, 20.6]	13.6
ColBench backend programming	colbench_text_deepseekv3	1	21.0	21.0 [18.5, 23.5]	21.0
	colbench_backend_programming colben	2	24.5	24.5 [21.7, 27.1]	24.5
	colbench_text_gemmini2flash	3	32.9	32.9 [30.0, 35.8]	32.9
	colbench_backend_programming colben	4	33.8	33.3 [30.3, 36.0]	33.6
	colbench_text_gpt41	7	35.6	35.3 [32.3, 38.0]	35.6
	colbench_backend_programming colben	8	37.2	36.8 [33.8, 39.6]	37.1
	colbench_example_agent_gpt41	9	38.6	38.2 [35.3, 40.9]	38.5

Benchmark	System	Pool	Reported	Iterated bootstrap [95% CI]	Con
	colbench_backend_programming colben	10	41.2	41.1 [38.1, 44.1]	41.2
	Hal Generalist Agent (Sonnet3.7)	22	43.7	43.2 [40.2, 46.1]	43.6
ColBench frontend design	colbench_text_deepseekr1	1	74.9	74.9 [72.8, 77.0]	74.9
	colbench_text_deepseekv3	2	75.2	74.7 [72.7, 76.8]	74.1
	colbench_text_gemini	3	81.6	81.7 [79.8, 83.6]	81.6
	colbench_text_o4minimedium	8	82.9	82.7 [80.9, 84.4]	82.9
CORE-Bench hard	HAL Generalist Agent(Deepseek-V3-03	1	8.9	8.9 [0.5, 17.3]	8.9 |
	CORE-Agent	6	11.1	5.1 [-3.4, 12.2]	4.8 |
	HAL Generalist Agent(o4-mini-low)	8	15.6	10.4 [1.5, 19.3]	13.9
	CORE-Agent(Deepseek-V3.1)	9	20.0	16.0 [6.4, 26.4]	17.0
	HAL Generalist Agent(gpt4.1)	19	22.2	14.9 [5.3, 25.3]	13.3
	Core Agent	23	40.0	40.2 [25.7, 54.6]	40.0
	CORE-Agent(Claude-Sonnet-4-5-high)	49	44.4	36.7 [23.3, 50.0]	40.0
	Coreagent(claude-opus-4-5)	56	75.6	75.8 [64.7, 89.1]	75.6
GAIA	HAL Generalist Agent (deepseek-chat	1	22.4	22.4 [16.0, 28.8]	22.4
	HAL Generalist Agent (gemini-2.0-fl	2	32.7	32.9 [26.2, 40.1]	32.7
	HAL Generalist Agent (o3-mini-2025-	3	33.9	31.8 [25.0, 37.7]	29.5
	HAL Generalist Agent (o3-mini-2025-	5	46.7	46.7 [39.5, 54.1]	46.7
	HAL Generalist Agent (gpt-4.1-2025-	7	49.7	47.7 [41.0, 54.3]	48.2
	HAL Generalist Agent (o4-mini-2025-	8	54.5	52.9 [45.4, 59.3]	54.2
	HAL Generalist Agent (o4-mini-2025-	10	58.2	56.9 [49.2, 63.7]	58.1
	HAL Generalist Agent (gpt-5-2025-08	13	59.4	57.0 [50.1, 63.4]	54.7
	HAL Generalist Agent (claude-3-7-so	20	64.2	62.6 [55.2, 69.2]	64.0
	HAL Generalist Agent (claude-sonnet	23	74.5	74.6 [68.1, 81.4]	74.5
Online-Mind2Web	SeeAct(gemini-2.0-flash)	1	26.7	26.7 [21.7, 31.7]	26.7
	Browser-Use_test(gemini-2.0-flash)	2	29.0	28.1 [23.2, 32.8]	28.8
	SeeAct(claude-haiku-4-5-20251001)	3	32.7	32.1 [26.7, 37.0]	32.6
	SeeAct(claude-haiku-4-5-20251001_th	5	36.3	36.3 [30.6, 41.6]	36.3
	SeeAct(gemini-2.5-pro-preview-03-25)	6	39.7	39.0 [33.3, 44.3]	39.6
	SeeAct(gpt-5-2025-08-07_medium_reas	8	42.3	41.5 [36.0, 47.0]	42.2
SciCode	Scicode Zero Shot Agent (gemini-2.0	1	1.5	1.5 [-1.5, 4.6]	1.5 |
	Scicode Zero Shot Agent (deepseek-a	2	3.1	2.6 [-1.7, 6.0]	2.8 |
	Scicode Zero Shot Agent (o4-mini-20	3	9.2	9.3 [1.6, 15.5]	9.2 |
ScienceAgentBench	SAB Self-Debug gemini-2-0-flash	1	12.7	12.7 [6.2, 19.2]	12.7
	SAB Self-Debug DS-V3	2	15.7	14.9 [7.7, 21.4]	15.6
	SAB Self-Debug o4-mini low	3	27.5	27.4 [18.7, 35.4]	27.4
	SAB Self-Debug o3 medium	8	33.3	31.7 [22.7, 40.3]	33.2
SWE-bench Verified mini	hal_generalist_agent_gemini20flash	1	2.0	2.0 [-1.9, 5.9]	2.0 |
	My Agent(gemini/gemini-2.0-flash)	4	22.0	22.1 [10.0, 32.0]	22.0
	SWE-Agent(claude-3-7-sonnet-2025021	10	54.0	54.3 [40.2, 68.2]	54.0
	SWE-Agent(gpt-5-2025-08-07)	13	64.0	63.3 [50.3, 76.3]	64.0
	SWE-Agent(claude-sonnet-4-5-2025092	19	68.0	65.7 [53.7, 77.7]	67.7
	SWE-Agent(claude-sonnet-4-5-2025092	21	72.0	69.7 [57.7, 81.7]	71.9
tau-bench airline	HAL Generalist Agent (gemini-2.0-fl	1	22.0	22.0 [10.4, 33.6]	22.0
	Taubench ToolCalling (gemini-2.0-fl	2	28.0	24.3 [13.3, 35.3]	25.1
	TAU-bench FewShot (gemini-2.0-flash)	3	44.0	44.5 [30.9, 56.9]	44.0
	Taubench ToolCalling (o4-mini-2025-	6	56.0	54.4 [40.3, 68.3]	55.8
	TAU-bench FewShot (o4-mini-2025-04-	13	60.0	55.1 [43.3, 67.3]	57.1
	TauBench FewShot High Reasoning (cl	42	62.0	52.6 [40.1, 64.1]	53.6
	TauBench Few Shot High Reasoning (c	45	66.0	58.4 [47.0, 69.0]	63.8
USACO	usaco_episodic	1	39.1	39.1 [33.6, 44.6]	39.1
	usaco_episodic	4	45.0	45.1 [39.2, 50.6]	45.0
	USACO Episodic + Semantic (o4-mini-	5	58.0	57.9 [52.4, 63.5]	58.0
	USACO Episodic + Semantic (gpt-5-20	8	69.7	69.7 [64.5, 74.9]	69.7

S4. Runs left out

Table S23: Runs read from HAL's archives but left out of the audit.

Benchmark	Run	Reason
assistantbench	Assistantbench Browser Agent (gemini-2.0-fl	no logged model calls on 33 of 33 tasks (cost unknown)
assistantbench	Assistantbench Browser Agent (o3-2025-04-16)	outcome coverage 0.00
corebench_hard	Claude Code (claude-opus-4-1)	no logged model calls on 45 of 45 tasks (cost unknown)
corebench_hard	Claude Code (claude-opus-4-5)	no logged model calls on 45 of 45 tasks (cost unknown)
corebench_hard	Claude Code (claude-sonnet-4-20250514)	no logged model calls on 45 of 45 tasks (cost unknown)
corebench_hard	Claude Code (claude-sonnet-4-5-20250929)	no logged model calls on 45 of 45 tasks (cost unknown)
corebench_hard	CORE-Agent	no logged model calls on 45 of 45 tasks (cost unknown)
corebench_hard	HAL Generalist Agent	no logged model calls on 33 of 45 tasks (cost unknown)
gaia	HAL Generalist Agent (o3-2025-04-16)	no logged model calls on 165 of 165 tasks (cost unknown)
online_mind2web	Browser-Use_test(claude-haiku-4-5-20251001_	outcome coverage 0.88
online_mind2web	Browser-Use_test(claude-haiku-4-5-20251001)	outcome coverage 0.88
online_mind2web	Browser-Use_test(claude-sonnet-4-5-20250929	outcome coverage 0.88
online_mind2web	Browser-Use_test(claude-sonnet-4-5-20250929)	outcome coverage 0.88
swebench_verified_mini	My Agent(o3-mini)	outcome coverage 0.92
swebench_verified_mini	My Agent(o3-mini)	outcome coverage 0.92
swebench_verified_mini	My Agent(o3-2025-04-16)	outcome coverage 0.68
swebench_verified_mini	My Agent(together_ai/deepseek-ai/DeepSeek-R	outcome coverage 0.02
swebench_verified_mini	My Agent(together_ai/deepseek-ai/DeepSeek-V	outcome coverage 0.62
swebench_verified_mini	SWE-Agent(o3-2025-04-16)	outcome coverage 0.94

S5. HAL's double counting of nested spans

HAL's reported token totals sum the usage block of every Weave span, so a provider call logged inside a wrapper span is counted twice. The table gives, per benchmark, how many of the runs read have a ratio of reported to once-counted tokens of exactly two, between one and two, and one.

Table S24: HAL's token total divided by the total with each model call counted once, per run. A ratio of exactly 2 means every provider call was logged inside a wrapper span that carried the same usage block.

Benchmark	Ratio 1	Between 1 and 2	Exactly 2
assistantbench	8	17	8
colbench_backend_programming	17	5	0
colbench_frontend_design	8	6	0
corebench_hard	33	2	26
gaia	12	6	18
scicode	17	11	9
scienceagentbench	13	0	12
swebench_verified_mini	21	0	24
taubench_airline	0	14	33
usaco	1	0	11
All	130	61	141

S6. Price table

Table S25: The price table used for every run (HAL's current table, dollars per million tokens). Cache reads and writes are priced at the cache rate where one is listed and at the input rate otherwise.

Model	Input ($/M)	Output ($/M)	Cache read ($/M)
Meta-Llama-3-1-70B-Instruct-htzs	2.68	3.54	
Meta-Llama-3-1-8B-Instruct-nwxcg	0.3	0.61	
Mistral-large-ygkys	4	12	
Mistral-small-zgjes	1	3	
amazon.nova-lite-v1:0	0.06	0.24	
amazon.nova-micro-v1:0	0.035	0.14	
amazon.nova-pro-v1:0	0.8	3.2	
anthropic/claude-3-5-haiku-20241022	1	5	0.1
anthropic/claude-3-5-sonnet-20240620	3	15	
anthropic/claude-3-5-sonnet-20241022	3	15	
anthropic/claude-3-7-sonnet-20250219	3	15	0.3
anthropic/claude-3.7-sonnet	3	15	
anthropic/claude-3.7-sonnet:thinking	3	15	
anthropic/claude-haiku-3.5	1	5	
anthropic/claude-opus-4	15	75	
anthropic/claude-opus-4-20250514	15	75	1.5
anthropic/claude-opus-4-5	5	25	0.5
anthropic/claude-opus-4-7	5	25	0.5
anthropic/claude-opus-4.1	15	75	
anthropic/claude-opus-4.1-20250805	15	75	1.5
anthropic/claude-opus-4.5	5	25	0.5
anthropic/claude-opus-4.7	5	25	0.5
anthropic/claude-sonnet-4	3	15	
anthropic/claude-sonnet-4-20250514	3	15	
anthropic/claude-sonnet-4-5	3	15	0.3
anthropic/claude-sonnet-4-5-20250929	3	15	
bedrock/amazon.nova-lite-v1:0	0.06	0.24	
bedrock/amazon.nova-micro-v1:0	0.035	0.14	
bedrock/amazon.nova-pro-v1:0	0.8	3.2	
bedrock/us.anthropic.claude-3-5-haiku-20241022-v1:0	0.8	4	
bedrock/us.anthropic.claude-3-5-sonnet-20241022-v2:0	3	15	
bedrock/us.anthropic.claude-3-opus-20240229-v1:0	15	75	
bedrock/us.anthropic.claude-3-sonnet-20240229-v1:0	3	15	
bedrock/us.meta.llama3-3-70b-instruct-v1:0	0.72	0.72	
claude-3-5-haiku-20241022	1	5	0.1
claude-3-5-sonnet-20240620	3	15	
claude-3-5-sonnet-20241022	3	15	
claude-3-7-sonnet-20250219	3	15	0.3
claude-haiku-3.5	1	5	
claude-haiku-4-5	1	5	0.1
claude-opus-4	15	75	
claude-opus-4-1-20250805	15	75	1.5
claude-opus-4-20250514	15	75	1.5
claude-opus-4-5	5	25	0.5
claude-opus-4-7	5	25	0.5
claude-opus-4.1	15	75	
claude-opus-4.1-20250805	15	75	1.5
claude-opus-4.5	5	25	0.5
claude-opus-4.7	5	25	0.5
claude-sonnet-4-20250514	3	15	
claude-sonnet-4-5	3	15	0.3
claude-sonnet-4-5-20250929	3	15	0.3
deepseek-ai/DeepSeek-R1	3	7	
deepseek-ai/DeepSeek-V3	1.25	1.25	

Model	Input ($/M)	Output ($/M)	Cache read ($/M)
deepseek/deepseek-chat-v3-0324	0.18	0.72	
deepseek/deepseek-chat-v3.1	0.27	1.1	
deepseek/deepseek-r1-0528	0.18	0.72	
gemini-2.0-flash	0.1	0.4	0.025
gemini-2.5-flash	0.5	2	0.05
gemini-2.5-pro	1.25	10	0.125
gemini-2.5-pro-preview-03-25	1.25	10	
gemini-3-flash-preview	0.5	3	0.05
gemini-3-pro-preview	2	12	0.2
gemini-3.1-pro-preview	2	12	0.2
gemini-3.5-flash	1.5	9	0.15
gemini/gemini-1.5-flash	0.075	0.3	
gemini/gemini-1.5-flash-latest	0.075	0.3	
gemini/gemini-1.5-pro	1.25	5	
gemini/gemini-2.0-flash	0.1	0.4	0.025
gemini/gemini-2.5-flash	0.5	2	0.05
gemini/gemini-2.5-pro	1.25	10	0.125
gemini/gemini-2.5-pro-preview-03-25	1.25	10	
gemini/gemini-3-flash-preview	0.5	3	0.05
gemini/gemini-3-pro-preview	2	12	0.2
gemini/gemini-3.1-pro-preview	2	12	0.2
gemini/gemini-3.5-flash	1.5	9	0.15
google/gemini-1.5-flash	0.075	0.3	
google/gemini-1.5-pro	1.25	5	
google/gemini-2.5-flash	0.5	2	0.05
google/gemini-2.5-pro	1.25	10	0.125
google/gemini-2.5-pro-preview-03-25	1.25	10	
google/gemini-3-flash-preview	0.5	3	0.05
google/gemini-3-pro-preview	2	12	0.2
google/gemini-3.1-pro-preview	2	12	0.2
google/gemini-3.5-flash	1.5	9	0.15
gpt-3.5-turbo	0.5	1.5	
gpt-3.5-turbo-0125	0.5	1.5	
gpt-4-turbo	10	30	
gpt-4-turbo-2024-04-09	10	30	
gpt-4.1-2025-04-14	2	8	0.5
gpt-4.1-mini-2025-04-14	0.4	1.6	
gpt-4.1-nano-2025-04-14	0.1	0.4	
gpt-4.5-preview-2025-02-27	75	150	
gpt-4o	2.5	10	
gpt-4o-2024-05-13	2.5	10	
gpt-4o-2024-08-06	2.5	10	
gpt-4o-2024-11-20	2.5	10	
gpt-4o-mini	0.15	0.6	
gpt-4o-mini-2024-07-18	0.15	0.6	
gpt-5	1.25	10	
gpt-5-2025-08-07	1.25	10	0.125
gpt-5-mini	0.25	2	
gpt-5-nano	0.05	0.4	
gpt-5-pro	15	120	
gpt-5.1	1.25	10	
gpt-5.2	1.75	14	0.18
gpt-5.2-2025-12-11	1.75	14	0.18
gpt-5.2-codex	1.75	14	0.18
gpt-5.2-pro	21	168	
gpt-5.4	2.5	15	0.25
gpt-5.4-mini	0.75	4.5	
gpt-5.4-nano	0.2	1.25	

Model	Input ($/M)	Output ($/M)	Cache read ($/M)
gpt-5.4-pro	30	180	3
gpt-5.5	5	30	0.5
gpt-5.5-2026-04-23	5	30	0.5
gpt-5.5-pro	30	180	3
meta-llama/Llama-4-Maverick-17B-128E-Instruct-FP8	0.27	0.85	
meta-llama/Meta-Llama-3.1-405B-Instruct	5	15	
meta-llama/Meta-Llama-3.1-70B-Instruct	0.88	0.88	
meta-llama/Meta-Llama-3.1-8B-Instruct	0.18	0.18	
o1-2024-12-17	15	60	
o1-mini-2024-09-12	3	12	
o1-preview-2024-09-12	15	60	
o3-2025-04-16	2	8	0.5
o3-mini-2025-01-31	1.1	4.4	0.55
o4-mini	1.1	4.4	
o4-mini-2025-04-16	1.1	4.4	0.275
openai/gpt-4.1-2025-04-14	2	8	0.5
openai/gpt-4.1-mini-2025-04-14	0.4	1.6	
openai/gpt-4.1-nano-2025-04-14	0.1	0.4	
openai/gpt-4.5-preview-2025-02-27	75	150	
openai/gpt-4o-2024-08-06	2.5	10	
openai/gpt-4o-2024-11-20	2.5	10	
openai/gpt-4o-mini	0.15	0.6	
openai/gpt-4o-mini-2024-07-18	0.15	0.6	
openai/gpt-5	1.25	10	
openai/gpt-5-2025-08-07	1.25	10	
openai/gpt-5-mini	0.25	2	
openai/gpt-5-nano	0.05	0.4	
openai/gpt-5-pro	15	120	
openai/gpt-5.1	1.25	10	
openai/gpt-5.2	1.75	14	0.18
openai/gpt-5.2-2025-12-11	1.75	14	0.18
openai/gpt-5.2-codex	1.75	14	0.18
openai/gpt-5.2-pro	21	168	
openai/gpt-5.4	2.5	15	0.25
openai/gpt-5.4-mini	0.75	4.5	
openai/gpt-5.4-nano	0.2	1.25	
openai/gpt-5.4-pro	30	180	3
openai/gpt-5.5	5	30	0.5
openai/gpt-5.5-2026-04-23	5	30	0.5
openai/gpt-5.5-pro	30	180	3
openai/gpt-oss-120b	0.15	0.6	
openai/o1-2024-12-17	15	60	
openai/o1-mini-2024-09-12	3	12	
openai/o1-preview-2024-09-12	15	60	
openai/o3-2025-04-16	2	8	0.5
openai/o3-mini-2025-01-31	1.1	4.4	0.55
openai/o4-mini-2025-04-16	1.1	4.4	0.275
openrouter/anthropic/claude-3-5-haiku	1	5	0.1
openrouter/anthropic/claude-3-7-sonnet-20250219	3	15	0.3
openrouter/anthropic/claude-3-haiku	0.25	1.25	0.03
openrouter/anthropic/claude-3.5-haiku	1	5	0.1
openrouter/anthropic/claude-3.7-sonnet	3	15	
openrouter/anthropic/claude-3.7-sonnet:thinking	3	15	
openrouter/anthropic/claude-opus-4	15	75	
openrouter/anthropic/claude-opus-4-20250514	15	75	
openrouter/anthropic/claude-opus-4.1	15	75	
openrouter/anthropic/claude-opus-4.1-20250805	15	75	
openrouter/anthropic/claude-opus-4.5	5	25	

Model	Input ($/M)	Output ($/M)	Cache read ($/M)
openrouter/anthropic/claude-sonnet-4	3	15	0.3
openrouter/anthropic/claude-sonnet-4.5	3	15	
openrouter/deepseek/deepseek-chat-v3-0324	0.18	0.72	
openrouter/deepseek/deepseek-chat-v3.1	0.27	1.1	
openrouter/deepseek/deepseek-r1-0528	0.18	0.72	
openrouter/google/gemini-2.0-flash-001	0.1	0.4	0.025
openrouter/google/gemini-2.5-flash	0.3	2.5	0.03
openrouter/google/gemini-2.5-pro	1.25	10	0.125
openrouter/google/gemini-3.1-pro-preview	2	12	0.2
openrouter/google/gemini-3.5-flash	1.5	9	0.15
openrouter/openai/gpt-oss-120b	0.15	0.6	
text-embedding-3-large	0.13	0	
text-embedding-3-small	0.02	0	
together/meta-llama/Meta-Llama-3.1-405B-Instruct-Turbo	3.5	3.5	
together/meta-llama/Meta-Llama-3.1-70B-Instruct	0.88	0.88	
together_ai/deepseek-ai/DeepSeek-R1	3	7	
together_ai/deepseek-ai/DeepSeek-V3	1.25	1.25	
us.anthropic.anthropic.claude-3-sonnet-20240229-v1:0	3	15	
us.anthropic.claude-3-5-haiku-20241022-v1:0	0.8	4	
us.anthropic.claude-3-5-sonnet-20240620-v1:0	3	15	
us.anthropic.claude-3-5-sonnet-20241022-v2:0	3	15	
us.anthropic.claude-3-opus-20240229-v1:0	15	75	
us.meta.llama3-3-70b-instruct-v1:0	0.72	0.72	